

DIGESTION

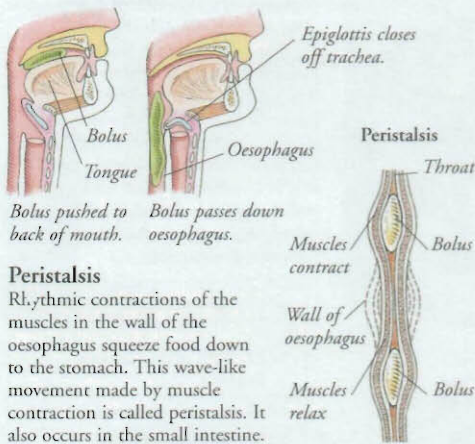
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THE BODY NEEDS THE nutrients in food to grow, maintain its structure, and provide energy. But the food we eat cannot be used by the body until it is processed by the digestive system. This is essentially a long tube, running from mouth to anus. As food passes along the digestive system it is chewed, and crushed, and then broken down chemically by enzymes. As it passes along the small intestine, food resembles a thin soup, and simple food molecules can be absorbed into the body itself by way of the bloodstream.

Swallowing

Once food is chewed, the tongue pushes the ball of food, or bolus, to the back of the mouth. As it touches the throat, the bolus triggers a reflex action and passes into the oesophagus. A flap called the epiglottis closes the entrance to the trachea (windpipe) to stop food entering the lungs.

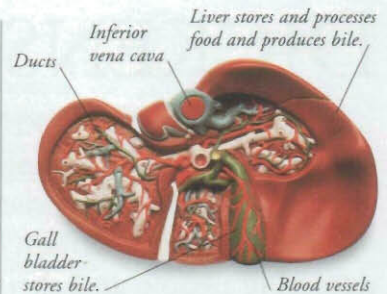
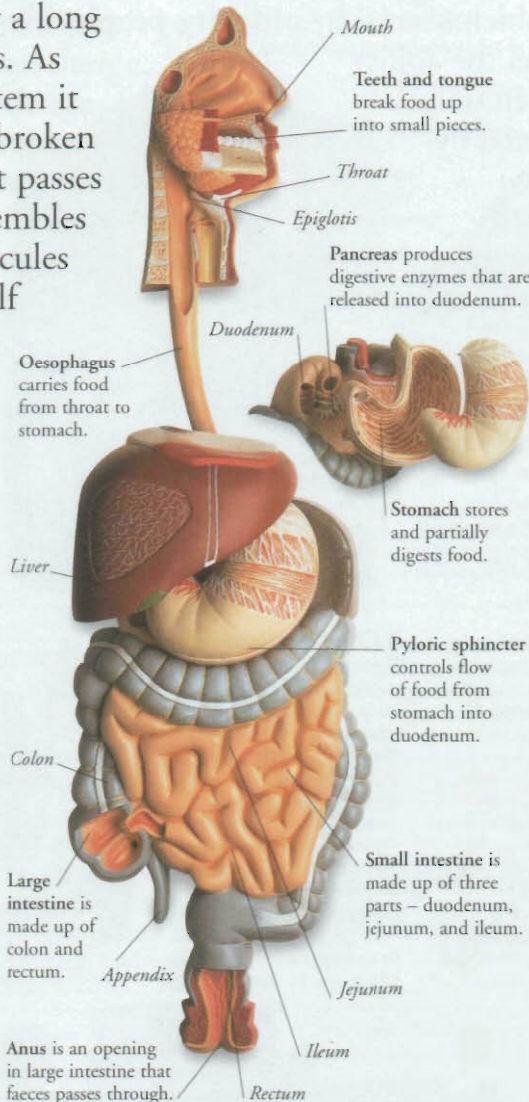


Peristalsis

Rhythmic contractions of the muscles in the wall of the oesophagus squeeze food down to the stomach. This wave-like movement made by muscle contraction is called peristalsis. It also occurs in the small intestine.

Digestive process

The digestive process has four stages: ingestion, digestion, absorption, and egestion. Ingestion happens when you eat food and is followed by digestion. Absorption is the transfer of food molecules into the bloodstream and egestion is the removal of waste as faeces.

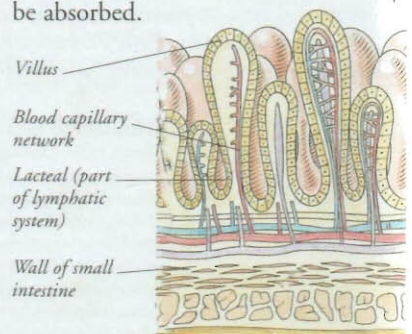


Liver, pancreas, and gall bladder

These three organs take part in digestion even though, since they have other body functions, they are not part of the digestive system. The liver produces bile, which is stored in the gall bladder and helps digest fats. The pancreas produces digestive enzymes that are released into the small intestine.

Absorption

Simple food molecules are absorbed into the bloodstream across the wall of the small intestine. Tiny finger-like projections called villi (singular: villus) greatly increase the surface area over which food can be absorbed.



Imaging the intestine

A special liquid is introduced into the large intestine to show clearly its position and internal shape. This type of X-ray enables doctors to detect signs of disease inside the large intestine without having to operate.

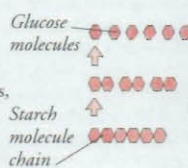


Food and enzymes

Enzymes are biological catalysts that speed up the conversion of one substance into another. Digestive enzymes speed up the breakdown of the complex carbohydrates, fats, and proteins that make up most of our food.

Carbohydrates

The body's main fuel, carbohydrate, comes in the form of sugars and complex carbohydrates, which include starch. Enzymes break starchy foods down into sugars such as glucose.



Fats

Fats provide the body with energy. Foods rich in fats include eggs and meat. Fats are broken down by enzymes in the small intestine to form fatty acids.

Proteins

Proteins are needed for growth and maintaining the body. Protein-rich foods are meat, fish, and nuts. Proteins are broken down into amino acids.

William Beaumont

The US Army surgeon William Beaumont (1785–1853) was the first to observe how food was digested in the stomach. In 1822, Beaumont treated a patient who had shot himself in the side and was left with an opening into his stomach. Through this opening, Beaumont was able to observe the stomach's movements during digestion and to record his findings.



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